Feature Engineering

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1. Baseline Model

Notebook: `Baseline_model.ipynb`
Features: all 432 raw columns

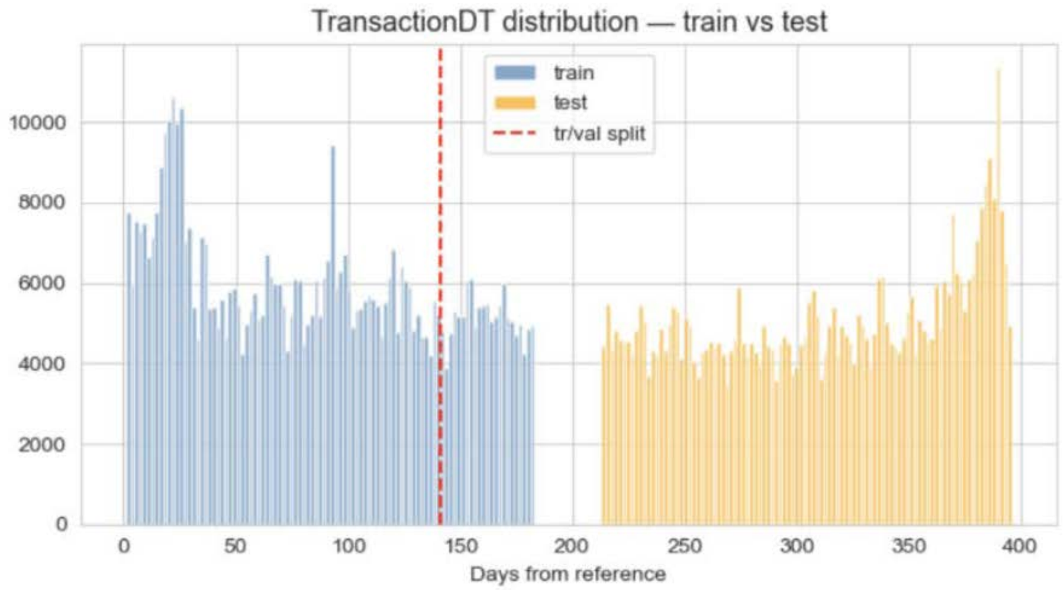
Categorical Handling

- All 50 categorical features encoded via `LabelEncoder` → integer
- Unknown values in val/test mapped to `'unknown'` before encoding
- No `categorical_feature` parameter passed to LightGBM — all features treated as numeric

> ⚠️ Label encoding only was used to establish a quick baseline. It is not appropriate for production use on categoricals features, cause absolute value doesn't mean anything.

Train / Validation Split

Time-based split on `TransactionDT`:



Split	Rows	Fraud rate	ratio
Train	472,432	3.51%	80%
Val	118,108	3.44%	20%

Distribution Shift Check

Key features with unknown rates in test vs val:

Feature	Val unknown %	Test unknown %
id_31	6.83%	22.06%
id_13	3.10%	13.74%
id_30	0.12%	4.39%
DeviceInfo	0.32%	3.39%
card1	1.09%	2.22%

Unknown rates are within acceptable range — no features dropped.

Model Config

```
n_estimators   = 500
learning_rate  = 0.05
num_leaves     = 256
early_stopping = 50 rounds
best_iteration = 207
```

Performance

Metric	Score
Val AUC	0.9208
Public LB	0.9229
Private LB	0.8974

> Private LB degrades relative to public LB, suggesting the model does not generalise well to the far-future test period.

Top 10 Features

Rank	Feature	Importance
1	card1	3508
2	TransactionDT	3222
3	TransactionAmt	2737
4	card2	2487
5	addr1	2299
6	D15	1340

7	dist1	1278
8	P_emaildomain	1166
9	C13	1025
10	card5	946

Zero Importance Features (removed list)

21 features with zero importance removed and added to `delete_cols`:

`id_29`, `V191`, `V196`, `V28`, `V241`, `V107`, `V117`, `V119`, `V120`, `V113`, `V305`, `V88`, `V89`, `V27`, `V41`, `V65`, `V325`, `V327`, `V68`, `id_16`, `id_27`

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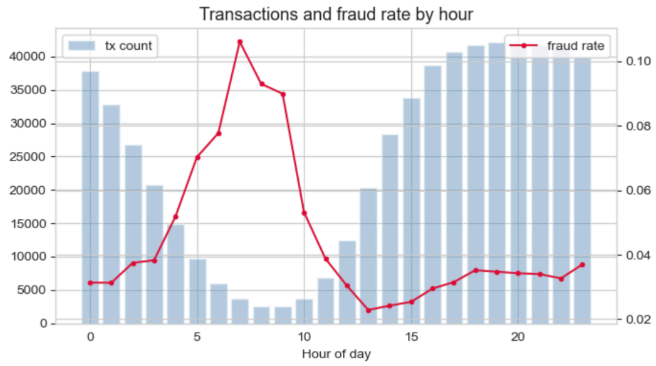
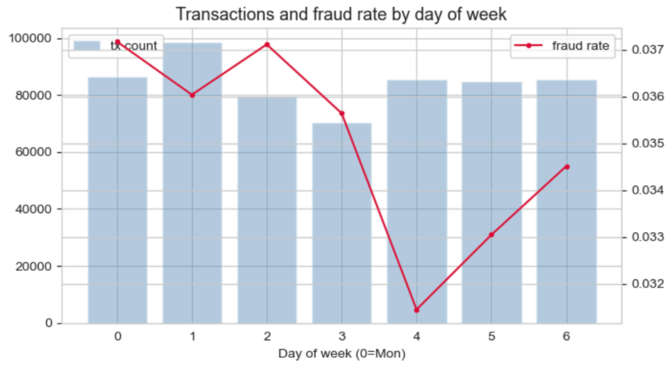
2. Improved Model — Encoding Experiments

Notebook: `model_v4.ipynb`

Engineered Features

- **TransactionDT Decomposition**



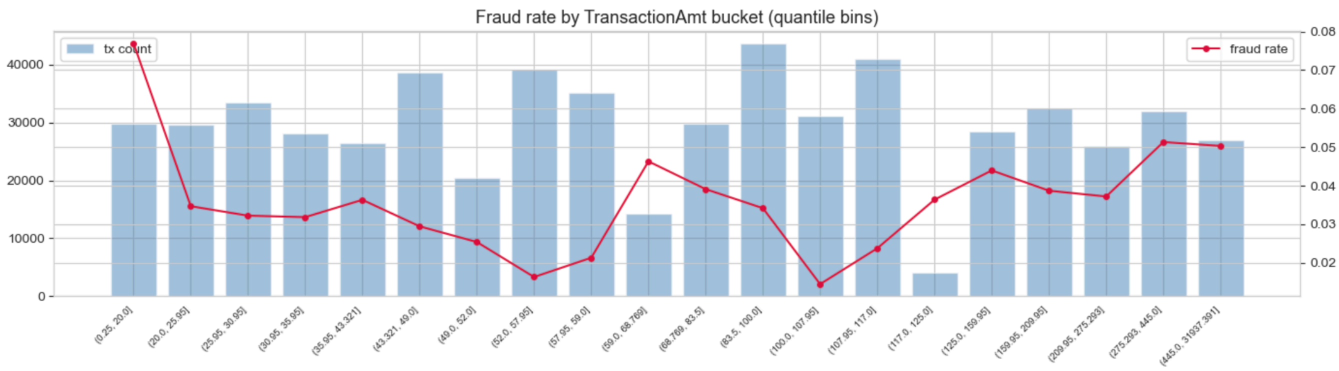


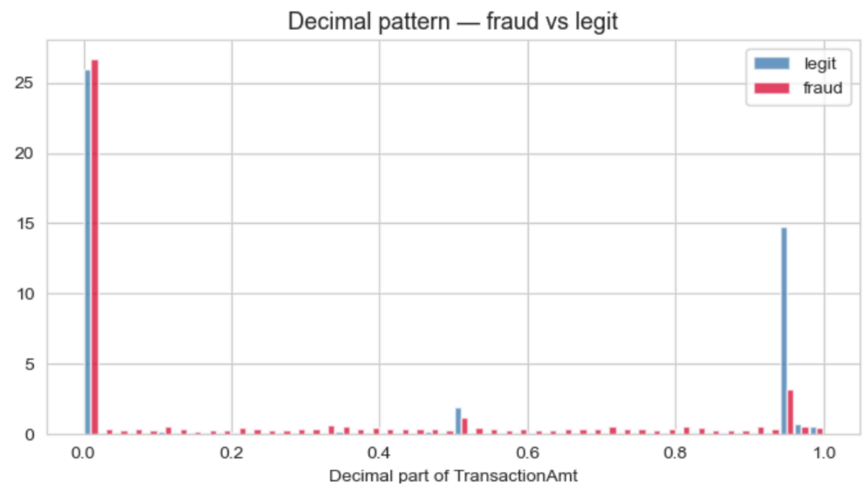
A clear spike in fraud rate during late-night and early-morning hours—unusual transaction period while fraud rate varies modestly by the day of week.

Feature	Description
`hour`	hour of day (0–23)
`dayofweek`	day of week (0–6)
`day`	absolute day index
`hour_sin`, `hour_cos`	cyclic encoding, period = 24
`dayofweek_sin`, `dayofweek_cos`	cyclic encoding, period = 7

Here we have sin/cos, encoding is used to represent cyclical time — **the model can learn that hour 23 and hour 0 are adjacent.**

- **TransactionAmt Features**





The first chart indicates that fraud rate fluactuates across all transaction amount ranges, But slightly higher fraud rate among smaller amount is still noticeble.

The second chart reveals that an abvious concentration of legit transaction scan be observed near amount ended with .99, which perhaps reflect the comon selling strategies, setting the price be ended with 0.99 instead of round number.

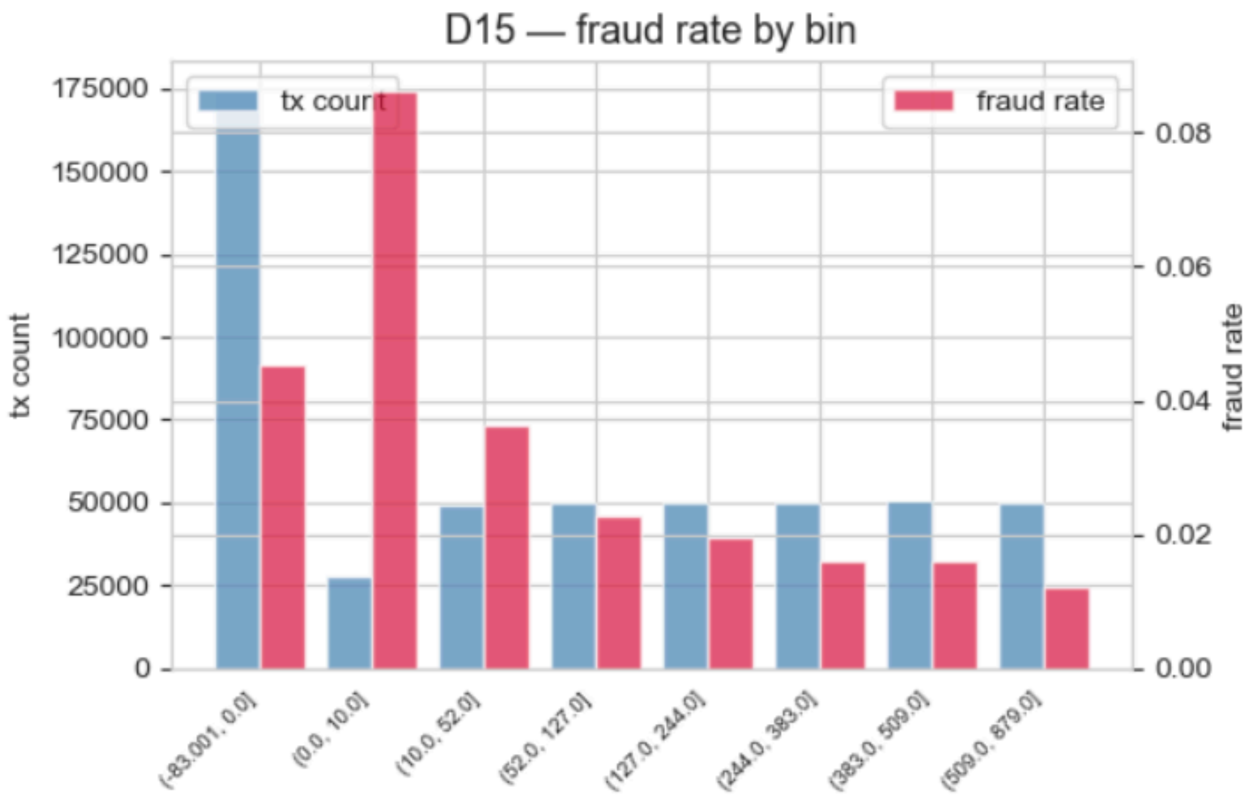
Feature	Description
`log_amt`	log1p of raw amount
`amt_is_round`	1 if amount has no decimal component
`amt_decimal`	decimal part rounded to 2 places
`amt_last_digit`	last digit for round amounts; −1 for non-round

BTW, `amt_decimal` became the single strongest feature across all model versions.

- **Log Transform for High-Skew Numerics**

1. Skewness computed on train only
2. All numeric features with `|skew| > 10` receive an additional `col_log = log1p(clip(col, 0))` column
3. Original column retained to keep original infor; 195 log columns added in total

- **D15 Binning**



`D15` replaced with `D15_bin`: 10 quantile bins computed on train, applied to val/test as a categorical feature.

Categorical Encoding Experiments

Experiment 1 — High-Cardinality Encoding Strategy

Strategies tested for features with > 255 unique values:

Strategy	A	B	C	label_encode
count freq	✅	✅	✅	❌
label encode	❌	❌	✅	✅
original value	✅	❌	❌	❌
Performance				
Val AUC	0.9061	0.9206	0.9257	0.9208
Public LB AUC	0.9021	0.9249	0.9274	0.9229
Private LB AUC	0.8799	**0.9014**	0.9010	0.8974

Key findings:

1. Passing high-cardinality features as native LightGBM categoricals triggers the internal 255-bin limit, severely degrading performance (−0.0147 AUC).
2. Frequency encoding alone (`B`) without label encoding achieve higher private score.
3. Combining label encoding and frequency encoding (`C`) gives higer score on Public, but fail at Private.

Experiment 2 — Sliding Window vs Global Aggregation

Aggregation features computed on `TransactionAmt` grouped by user/card keys:

Method	Aggregations	Val AUC	Public LB	Private LB
Sliding window	cnt, avg, min, max	0.9156	0.9165	0.8953
Global	cnt, avg, min, max, p25, p50, p75	**0.9310**	**0.9310**	**0.9062**

Global aggregation outperforms sliding window despite **being less realistic for production:**

- Both train and test benefit from future information in the full dataset, also it could not happen in real situation.
- card overlap exists between train and test, making global statistics informative

Feature importance

feature	importance
amt_decimal	2279
TransactionAmt	1061
TransactionDT	1012
id_31	635
C13	582
card1_dist1_avg	569
D15	521
card1_C13_avg	511
card1_TransactionAmt_max	484
dist1	480
card1_dist1_max	460
card2_TransactionAmt_max	455
card1_TransactionAmt_avg	439
card1_C13_max	433
card1_dist1_p75	430

Final Performance

Metric	Score
Val AUC	**0.9310**
Public LB	0.9310
Private LB	0.9062

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3. User ID Construction

Notebook: `model_v5.ipynb`

Aggregation features are only meaningful when grouped by the correct user/cardID identity. Without a true identity, aggregating conflates multiple users, diluting fraud signals with legitimate transaction history.

> Aggregation features measure deviation from a user's historical behaviour. The signal is lost if transactions from different users are mixed in the same group.

Candidate Formulas

Based on Chris Deotte's approach, user identity is approximated by combining card and address features:

ID	Formula	Distinct count	Mean tx/user	p50	p75
user_id1	card1 + addr1 + D1_norm	217,850	2.71	1	3
user_id2	card1 + card2	14,525	40	4	13
user_id3	card1 + addr1	39,974	14	2	7

user_id4	card1 + addr1 + P_emaildomain	90,375	6	2	4

`user_id1` is the most granular (fine-grained) but too sparse — most users have only 1 transaction per half-year.

Experiment 1 — Individual ID Selection

user_id	LogLoss	Val AUC
baseline	0.08139	0.9310
user_id1	**0.07850**	**0.9362**
user_id2	0.08215	0.9293
user_id3	0.08163	0.9325
user_id4	0.08206	0.9324

I conducted aggregation on multiple candidate user_id. The result indicates `user_id2` is eliminated. Remaining IDs tested in combination:

1, user_id1 + user_id3+ user_id4==> AUC 0.9389

2 ,user_id1 + user_id3 ==> AUC 0.9390.

3. user_id1 + user_id4 ==> AUC 0.9364

Final Performance

Metric	Score
Val AUC	**0.9390**
Public LB	0.9358
Private LB	**0.9110**

4. Categorical Aggregation Features

Notebook: `model_v6.ipynb`

Beyond numeric aggregation, categorical behaviour features were constructed to capture consistency patterns within user groups:

Feature type	Description
`n_distinct_{col}`	number of distinct values seen for this user
`mode_cnt_{col}`	frequency of the most common value for this user
`is_mode_{col}`	1 if current transaction value matches the user's mode

Features constructed for: `card3`, `card4`, `card5`, `card6`, `addr1`, `addr2`, `R_emaildomain`

Feature Selection

Each of the 12*3 candidate features was added individually to the model and validated. Only features with positive AUC contribution were retained.

Feature	AUC	Delta
n_distinct_R_emaildomain	0.939193	+0.000193
n_distinct_card6	0.939068	+0.000068
mode_cnt_R_emaildomain	0.939066	+0.000066
n_distinct_addr1	0.939044	+0.000044
is_mode_card6	0.939044	+0.000044
n_distinct_card3	0.939044	+0.000044
is_mode_card3	0.939044	+0.000044

n_distinct_card4	0.939044	+0.000044
is_mode_card4	0.939044	+0.000044
is_mode_addr2	0.939044	+0.000044
is_mode_addr1	0.939044	+0.000044
is_mode_card5	0.939044	+0.000044
n_distinct_P_emaildomain	0.938965	-0.000035
mode_cnt_card2	0.938521	-0.000479
is_mode_DeviceType	0.938472	-0.000528
is_mode_card2	0.938358	-0.000642
mode_cnt_DeviceInfo	0.938181	-0.000819
mode_cnt_DeviceType	0.938181	-0.000819
is_mode_P_emaildomain	0.938082	-0.000918
n_distinct_card5	0.938042	-0.000958
mode_cnt_card5	0.937954	-0.001046
mode_cnt_addr2	0.937941	-0.001059
is_mode_R_emaildomain	0.937922	-0.001078
n_distinct_DeviceInfo	0.937712	-0.001288
mode_cnt_card4	0.937465	-0.001535
n_distinct_DeviceType	0.937395	-0.001605
n_distinct_addr2	0.937252	-0.001748
mode_cnt_card6	0.937230	-0.001770
mode_cnt_P_emaildomain	0.937156	-0.001844
n_distinct_card2	0.936951	-0.002049
mode_cnt_addr1	0.936860	-0.002140

mode_cnt_card3	0.936683	-0.002317
is_mode_DeviceInfo	0.935907	-0.003093

Final Performance (model_v6)

Metric	**Score**
Val AUC	**0.9397**
Public LB	0.9361
Private LB	**0.9099**